Additive Manufacturing with Robotics to Enable Swarm Applications

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Problem and Requirements

The manufacturing industry has changed rapidly in recent years. A transformation from inefficient, expensive, and resource-intensive processes to highly automated and streamlined. Technologies like robotics and 3D printing have been at the helm of this evolution.

In the past, manual labor and basic mechanical systems dominated the operations of manufacturing systems due to their adaptability and ease of use, however, they lacked efficiency and precision, valuable traits in an assembly line. Robotics introduced programmable, accurate, reliable, and independent machines that automated countless manufacturing processes. Robots could work continuously in demanding conditions while maintaining high quality and fast production, which enabled scalability.

Swarm manufacturing emerged out of the integration of robotics into manufacturing. Multiple robots working together to complete tasks and produce desired products fostered scalability, customization of processes, cost efficiency, autonomy, and increased yield (**Figure 1)**.

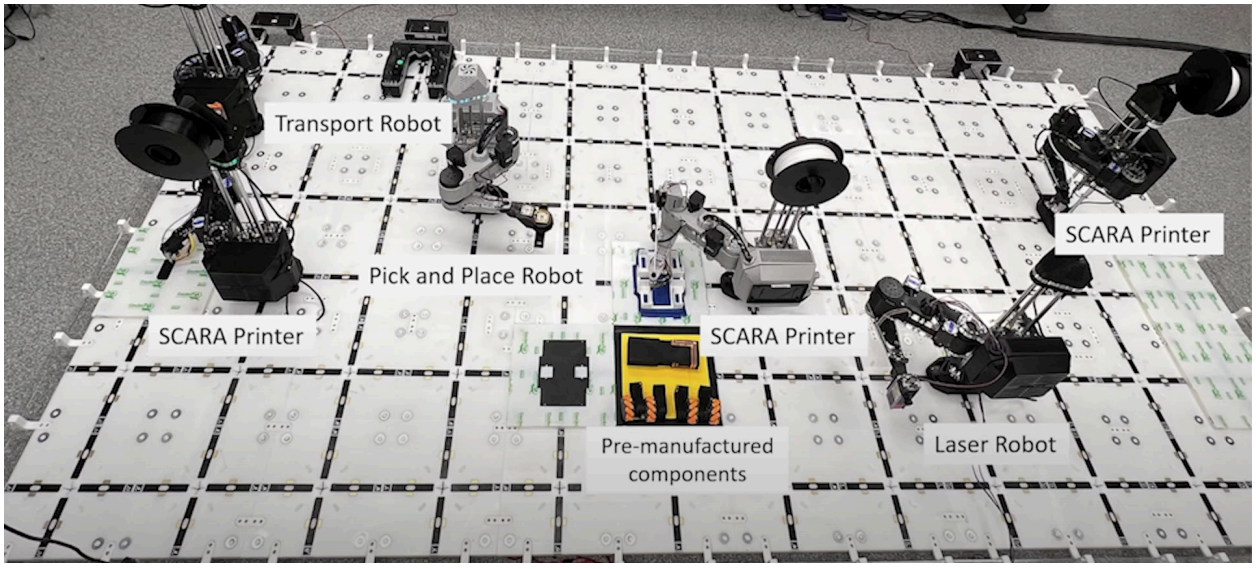


Figure 1: Swarm Manufacturing
Source: AMBOTS Inc.

3D printing, another cornerstone of the manufacturing revolution, allowed users to manufacture products with complex geometries in a fast, cost-effective, and resource-efficient manner. The most prevalent type of 3D printing is fused deposition modeling or FDM, due to its simplicity, higher speed compared to other 3D printing methods, and quality finish. In FDM 3D printing, melted plastic filaments are deposited through an extruder and hot end, one layer at a time, to create the product. FDM printers, generally, use a single nozzle to deposit the melted filament, which extends the process and limits the complexity of the product that can be printed.

Swarm manufacturing with robotics can be used to solve this problem. Multiple flexible robotic arms can work collaboratively to deposit filament in a structured way to optimize the FDM process. To date, there are no commercially available FDM extrusion systems for 6-degree-of-freedom robotics arms that are vital to implementing swarm manufacturing in the 3D printing industry. Such a system could be the next revolutionary step in the evolution of manufacturing. It will increase efficiency and provide an industrially scalable solution to the relatively small-scale scope of current 3D printing technologies. The most exciting avenue that swarm manufacturing will open up for 3D printing is conformal printing, printing in non-planar or 3D geometries, which will allow users to print complex shapes and overcome the design limitations of current 3D printing systems.

Project Goals

The primary objective of this project was to design a 3D printer extruder that would work with the existing infrastructure the lab had in place, that being the UR5e 6 DOF arm. This process involved working on the design of a new toolhead for the UR5e capable of depositing filament onto the bed print surface and integrating that mechanism onto the UR5e arm, all while

following the design engineering requirements set. The final deliverable of this project was a test print using the entire system, which would serve as a proof of concept that the designed extruder mechanism is functional and can be integrated with the UR5e arm to enable further swarm applications within additive manufacturing.

Functional Description

The entire project relies on multiplanar movement abilities, which are achieved with the UR5e arm. The UR5e arm **(Figure 2)** is a commercial arm built by Universal Robotics that has 6 degrees of freedom, enabling precise movements across multiple axes. The flexibility in movement makes it ideal for tasks requiring complex manipulation and positioning. In this project, the UR5e arm is responsible for accurately positioning the extruder mechanism, allowing for precise printing along programmed paths.

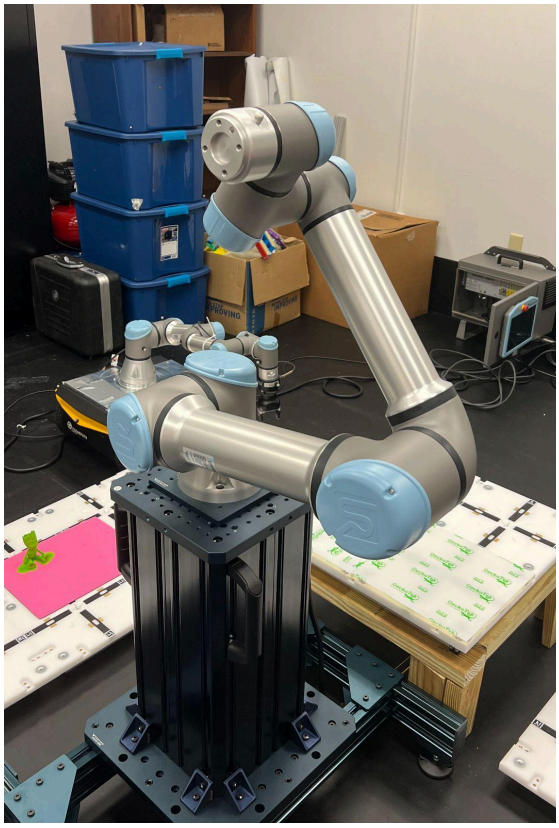


Figure 2: UR5e 6 DOF Robotic arm

This project began by looking at the customer requirements, ensuring that the final project could achieve most of the goals. Initial customer needs included that the final system be reconfigurable, have a higher feed rate than current industry machines (being greater than the existing Ambots swarm manufacturing robot), and be capable of conformal printing all while being easy to assemble and learn. Using these requirements, an analysis of the engineering requirements was performed, noticing that the final prototype should be under 5 kilograms (that being the max weight the UR5e can safely hold), should have a modular design, and should have a skinny hot-end, allowing the printer a maximum amount of rotational range for conformal printing.

Using Onshape CAD (Computer Aided Design), an initial prototype **(Figure 3)** was designed in 4 main parts, the tool holder, the tool head, the part cooling fans, and the spool holder.

The tool holder mechanism **(Figure 4)** was mounted directly to the end effector of the UR5e arm and served to allow multiple different types of extruder mechanisms to be slotted in making the overall design modular. This served as beneficial as the UR5e's original mounting points are oriented normal to the tool head and have a relatively wide profile which restricts our ability to have easily removable tool heads. The geometry of the UR5e side of our mount conforms to the robot's end effector and mounting holes and we utilized M6 bolts to fasten our mount to the robot. On the tool head side, we designed a rectangular slot that would correspond to a rectangular feature on the tool head. The two parts mate together and are secured by two M6 bolts that are screwed through both pieces. This particular design allows us to access the tool head's mounting fasteners from the side, giving us greater flexibility in the design while still

retaining the ability to quickly swap tool heads out. This also ensures that the extruder mechanism is held securely, allowing for consistency within test prints.

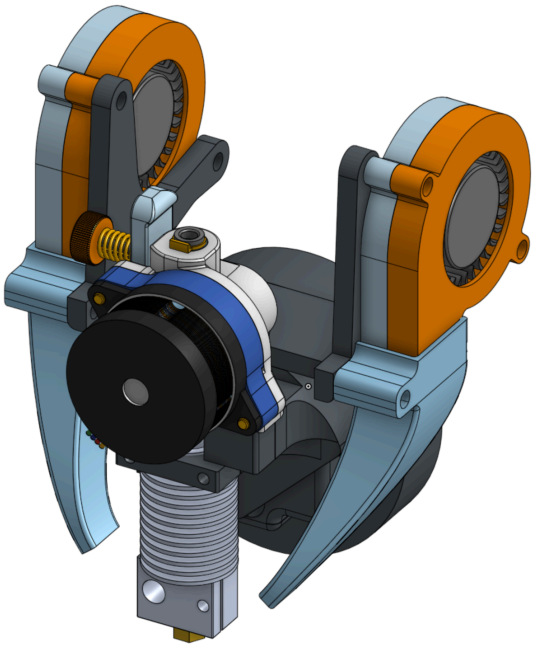


Figure 3: Fully Assembled System in CAD

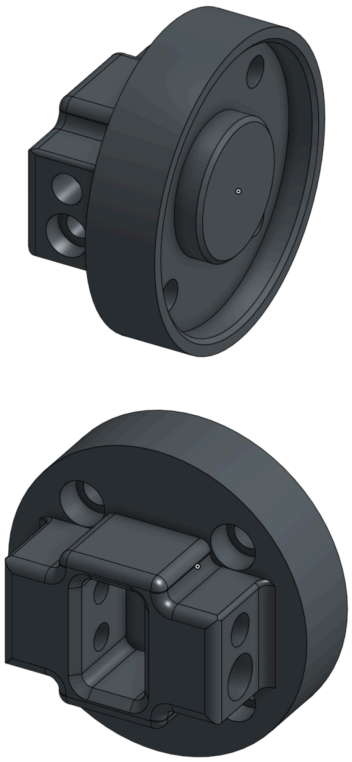


Figure 4: Tool Holder in Onshape

Another component of the design included the tool head **(Figure 5)** which was designed with the primary purpose of depositing hot filament onto the printing surface. To achieve this we used off-the-shelf components and created 3D models that effectively incorporate them. For the extruder, we used an Orbiter V2 Direct Driver extruder, as we found that it was more effective for the filament to be pulled into the system rather than pushed into the system as it would have been without a direct drive system. For the hot end, we used an E3D V6 Volcano Hotend and for the hot-end cooling fan, we used a small 30mm fan. The toolhead itself has a rectangular

extrusion on the face that interfaces with the tool holder, allowing it to be held securely in place. On the top of the tool head, there are spots for heat set inserts for the part cooling fans and mounting points for the direct drive extruder. On the front side of the tool head is a slot for the hot end to be inserted and mounting points for a clamp system, allowing the hot end to be mounted to the tool head.

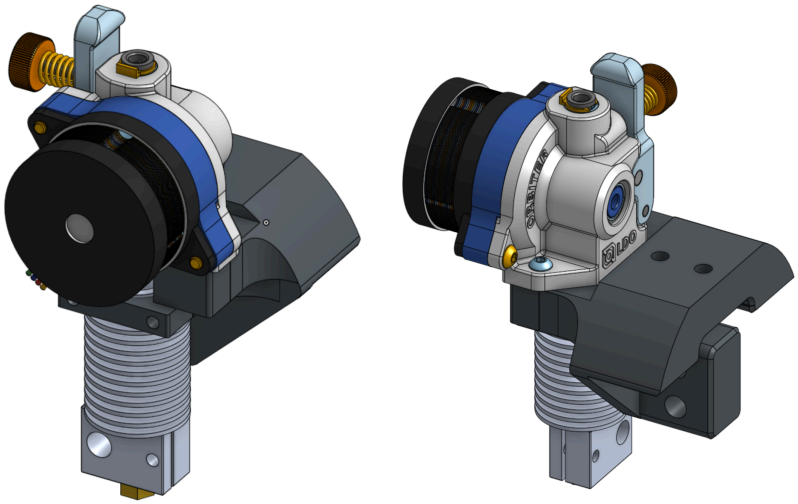


Figure 5: Toolhead in Onshape CAD

Additionally, part cooling fans **(Figure 6)** were designed to be mounted on top of the tool head, ensuring that air would flow down to the surface, quickly cooling down the part, aiding when printing on nonplanar surfaces or with the goal of conformal printing. This design consisted of two 5015 Radial Blower fans connected to “funnels” that would direct the air to the print surface. However, these parts did not make it into our final printing test as they were deemed irrelevant to the task at hand.

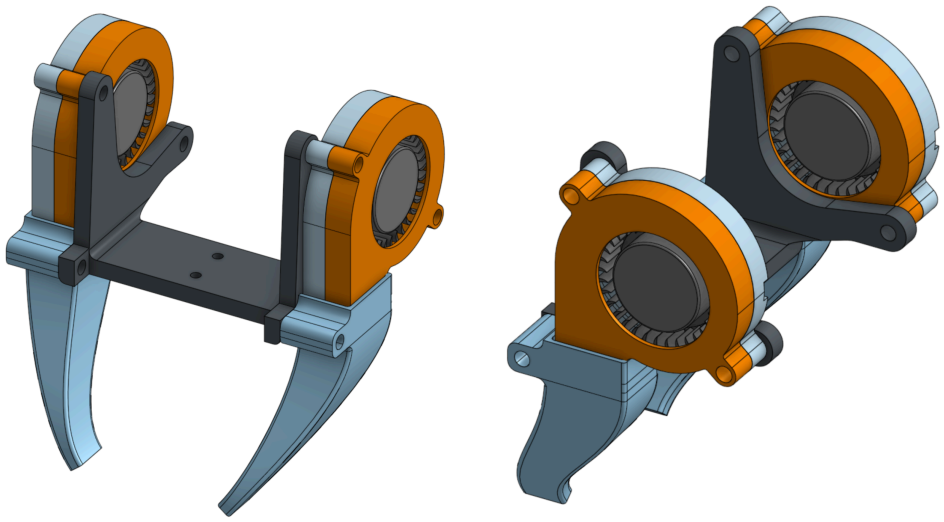


Figure 6: Part Cooling Fan Assembly

Lastly, we designed a filament routing system to direct filament to the direct drive extruder on the tool head. We decided to mount the filament spool to the first hub on the arm. Placing the filament at this location would allow the spool to rotate with the arm, ensuring that the large spool is located in the same position relative to the robot at all times. This is crucial for avoiding collisions when the arm does any complicated movements. Our design is a “hubcap” **(Figure 7)** that conforms to the existing geometry of the UR5e’s first hub. Our CAD model, while effective, is not as clean as it could be. This is due to invalid geometries while referencing the UR5e’s CAD model, which made creating a perfect conformed shape difficult within our time constraint. On the side of the hubcap, we mounted a filament spool holder. On top of the hubcap, we placed a mount for the PTFE tubing. The 1.75mm PTFE tube will run from the hubcap to the tool head, constraining the filament path. This keeps any filament from getting tangled up during the arm’s movements.

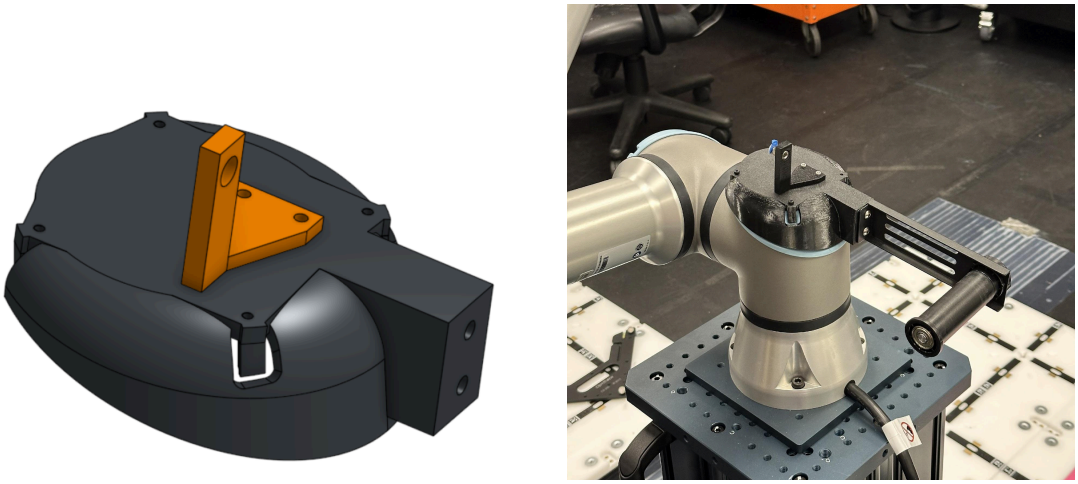


Figure 7: Filament Routing Assembly

All of these designs were then printed in PETG (Polyethylene Terephthalate Glycol) for their higher strength under load and higher heat resistance when compared to standard materials like PLA. After printing, we mounted all the parts to the UR5e robot to get our final assembly **(Figures 8 and 9).**

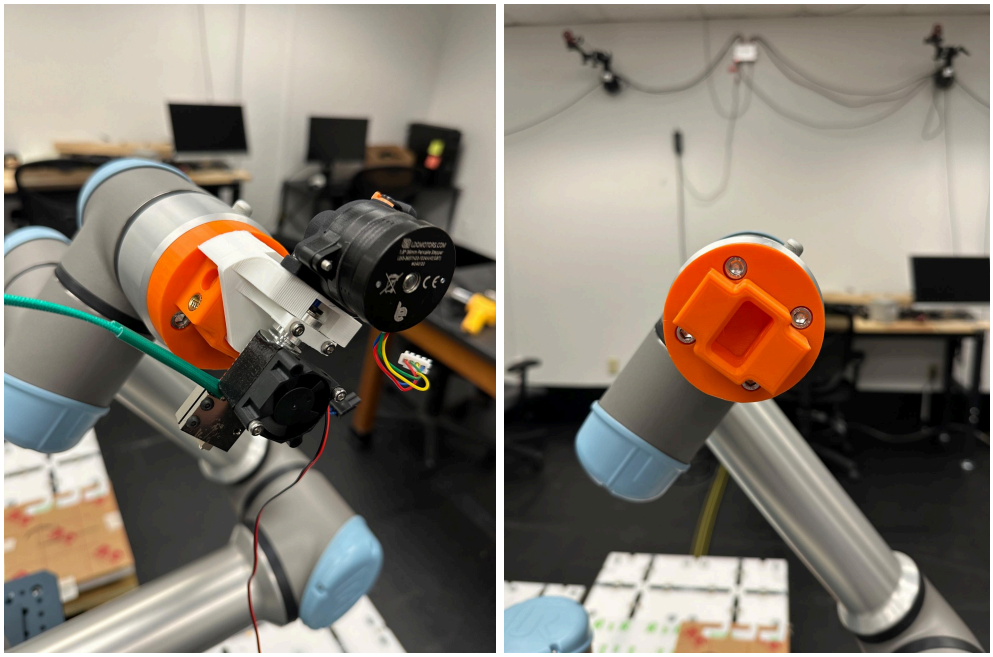


Figure 8: Assembled Extrusion Assembly

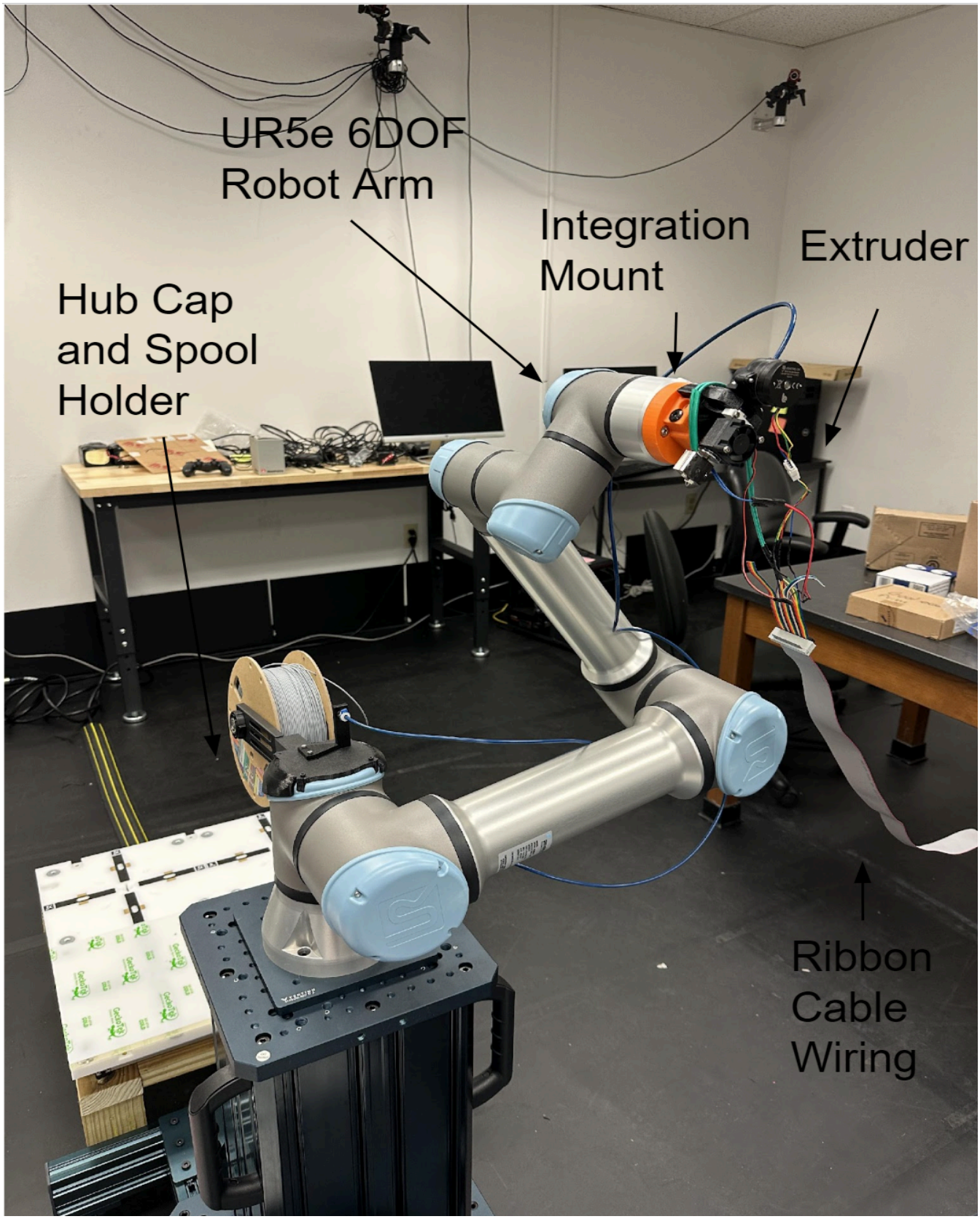


Figure 9: Labeled Full Robot Assembly

Another crucial part of the system we developed was the wiring. We chose to use ribbon cables **(Figure 9)**, as this would simplify the routing and keep us from having to use additional wire management in the form of wire looms. In order to keep the system as modular as possible, we used a long ribbon cable and crimp connectors on each end. Then we made two additional ribbon cables that would interface with the long ribbon cable on one end, and interface with either the Duet Board or toolhead on the other end. This worked very well to keep our wires managed. One consideration when using the ribbon cable that we had to address was the power draw of the hot end. The heating element drew too much current for a single wire, so we had to use eight of the 26 available wires in the ribbon cable for the heater.

To control the 3D printing side of the toolhead we used the Duet 2 wifi board **(Figure 10)**. While we did not do any particular work on the software side besides changing a few parameters, the board was extremely helpful in conducting our tests. The Duet allowed us to control hot end temperature, and extrusion rates, and power all of our components.

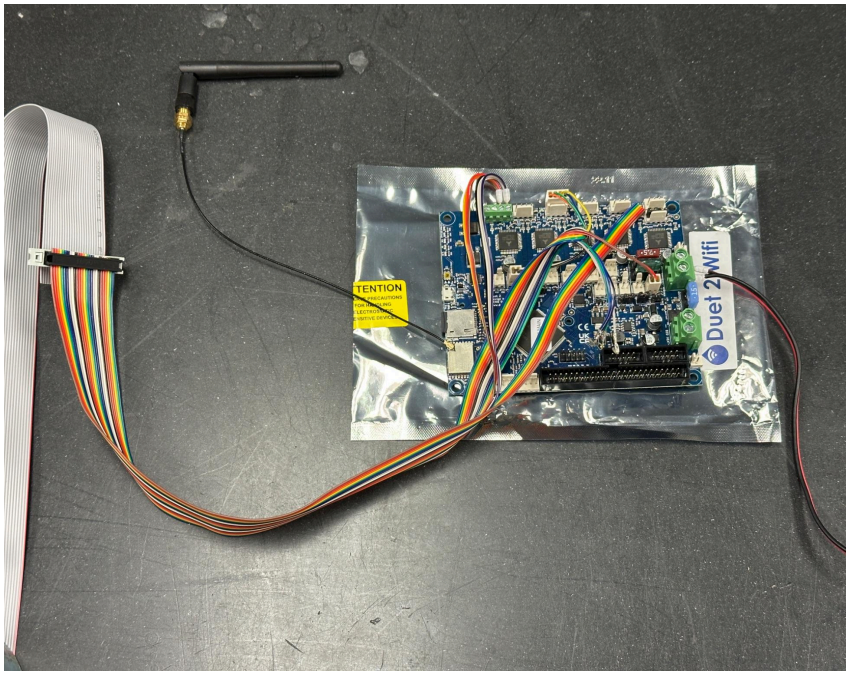


Figure 10: Ribbon cable wiring and Duet 2 Wifi Board

Results

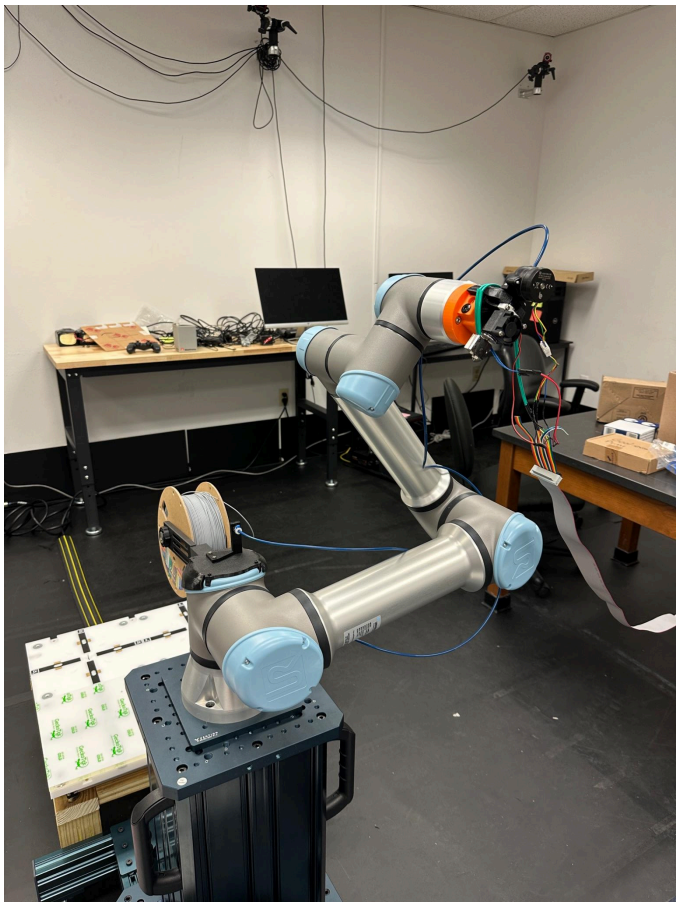


Figure 11: Fully Assembled System integrated with UR5e

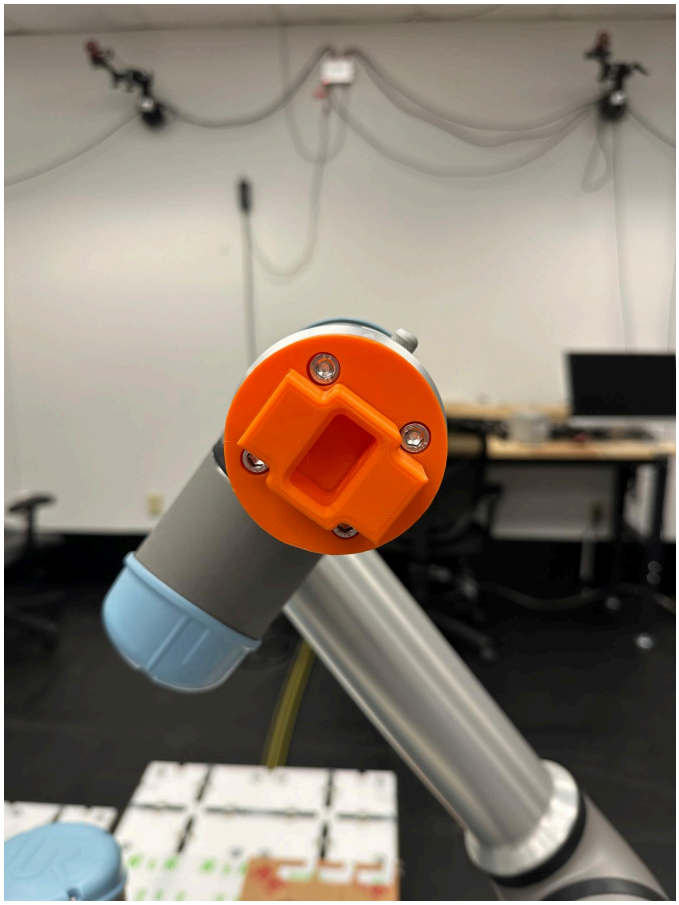


Figure 12: 3D Printed Tool Mount

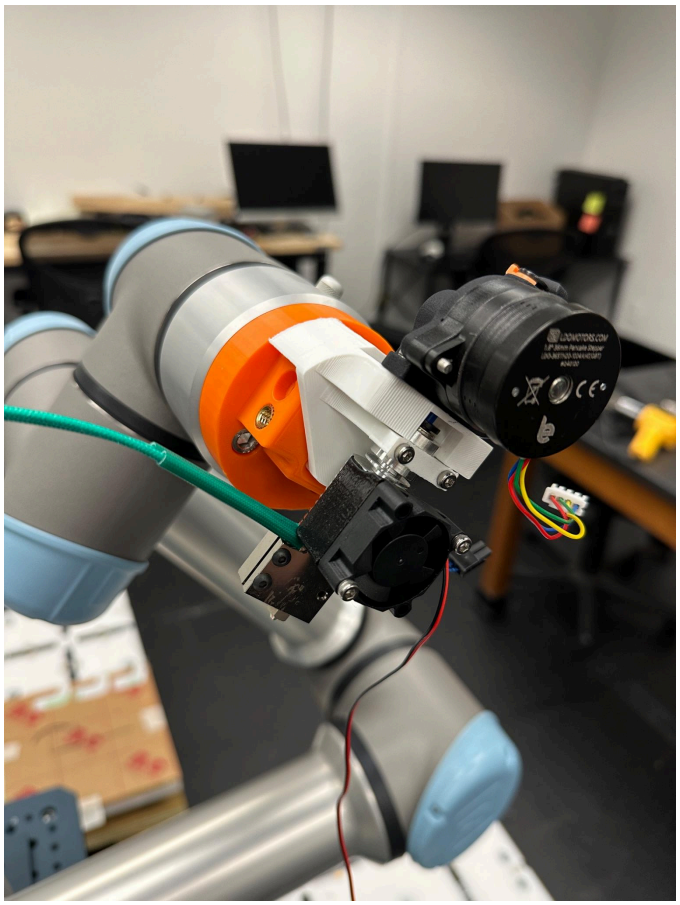


Figure 13: Removable Toolhead attached to 3D Printed Mount

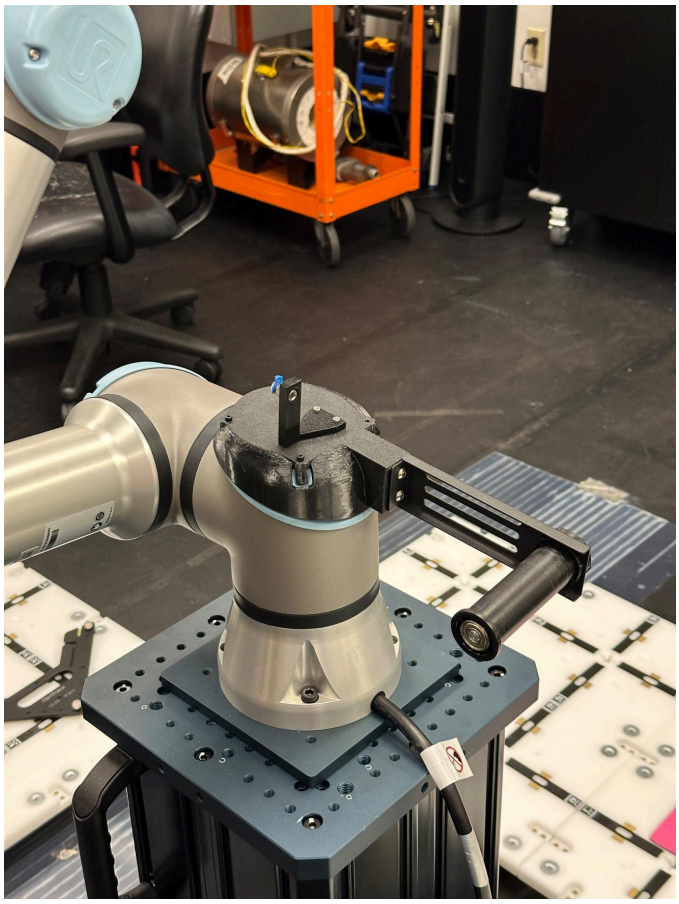


Figure 14: Hubcap and Filament Routing System

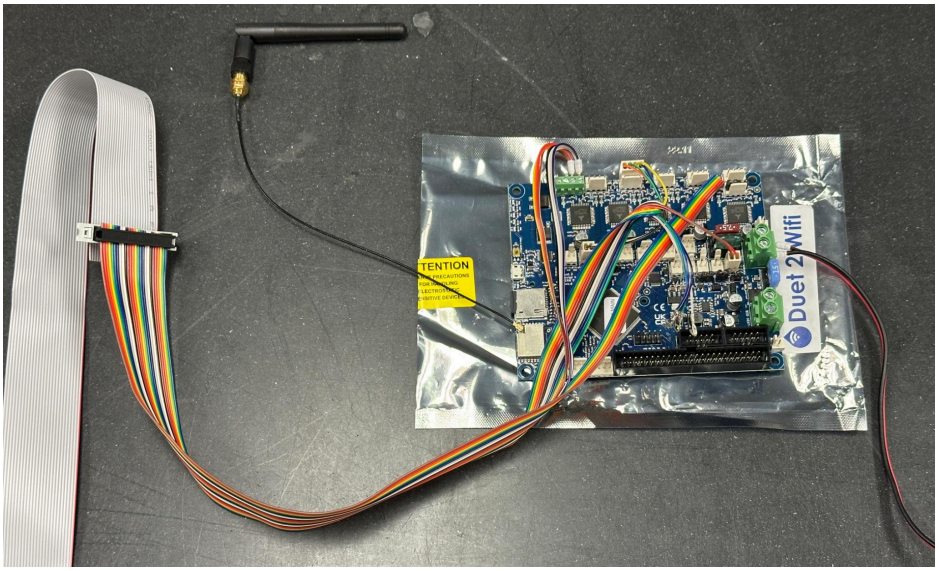


Figure 15: Ribbon cable wiring and Duet 2 Wifi Board

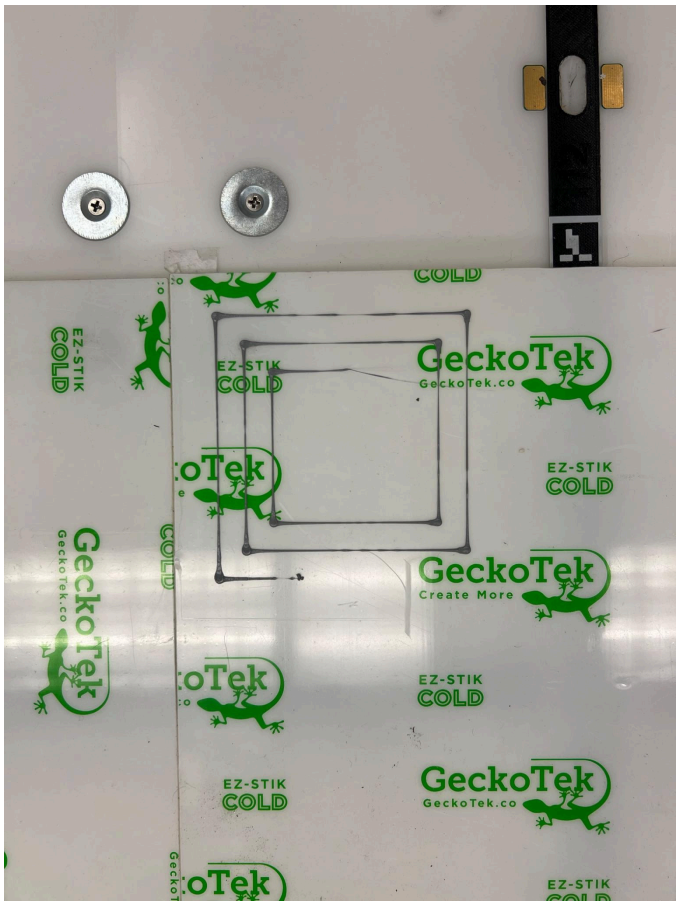


Figure 16: Gecko Tek surface and printed line

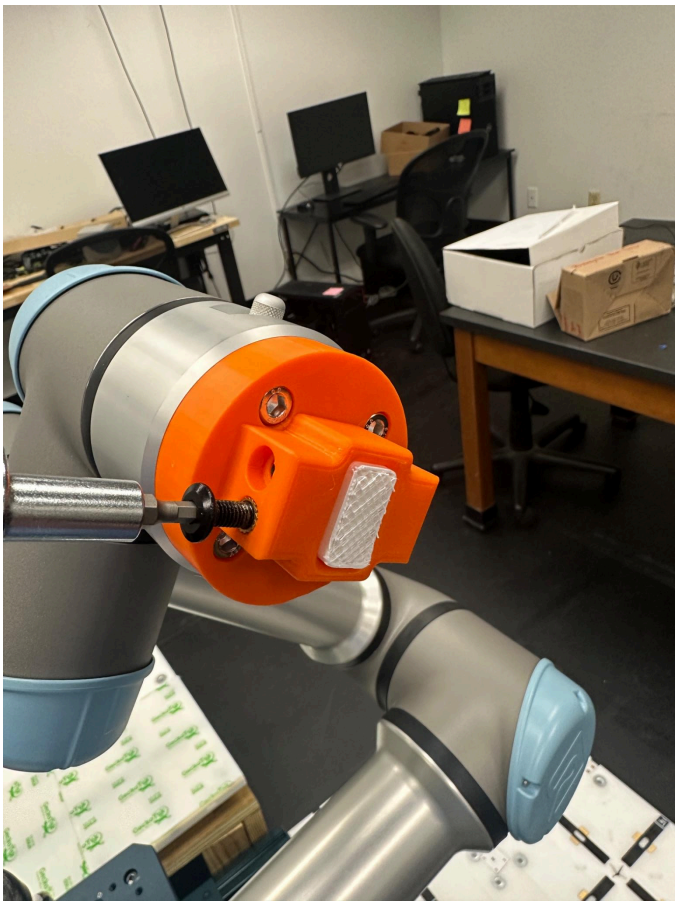


Figure 17: Toolhead Part failure after collision

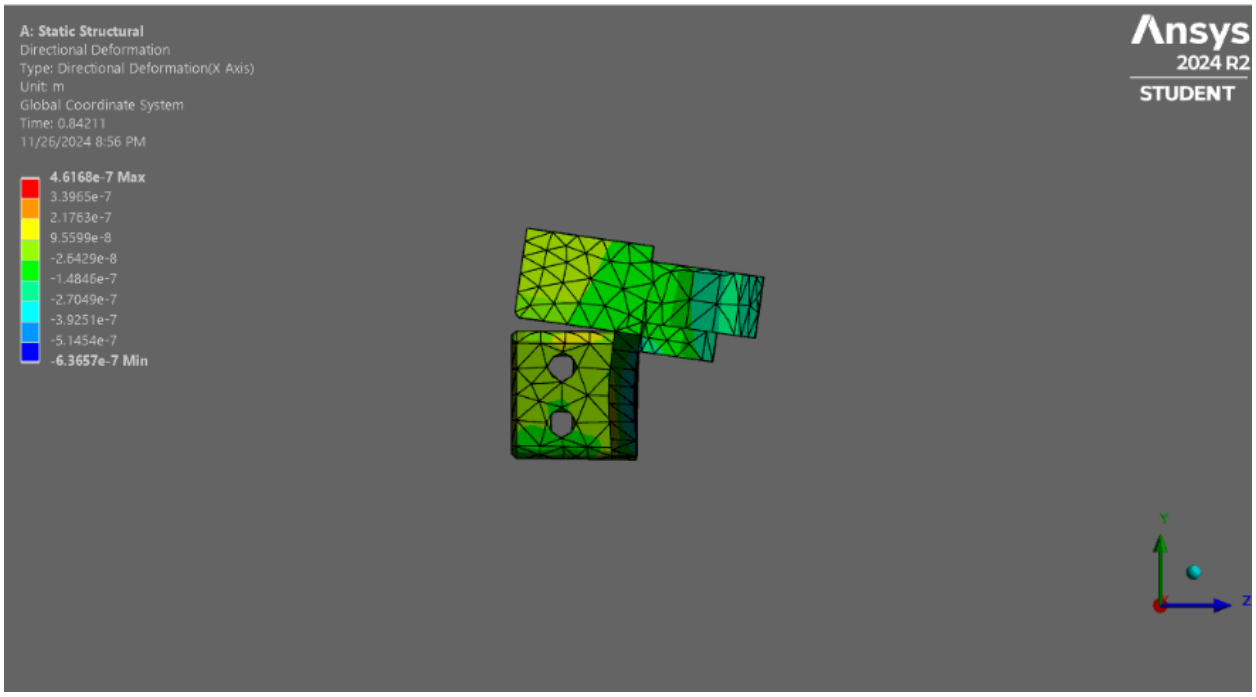


Figure 18: Toolhead part under load

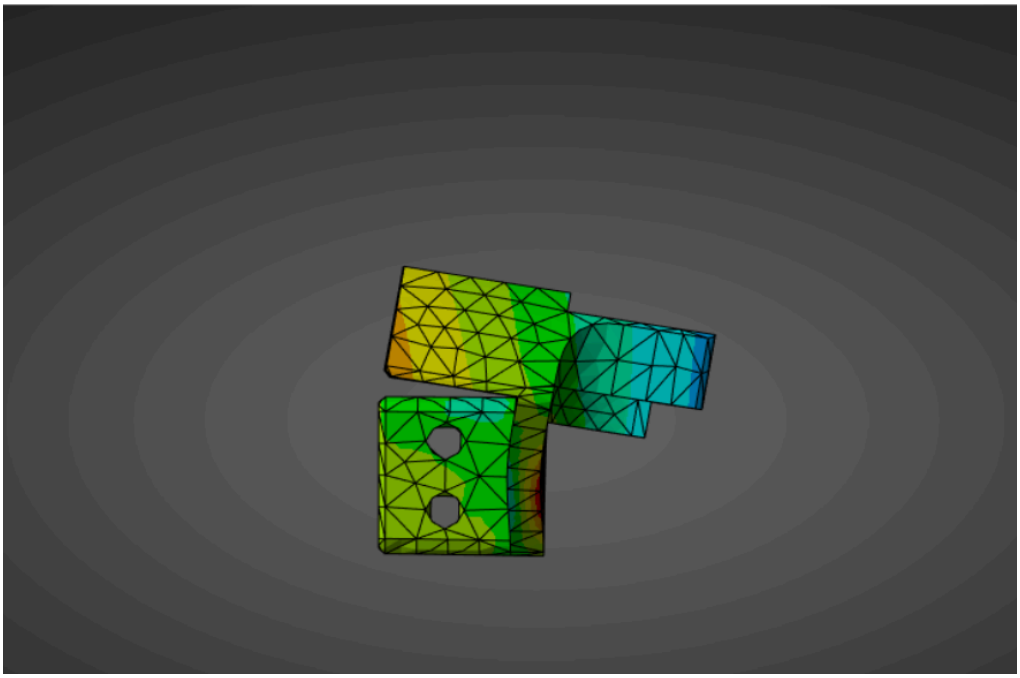


Figure 19: Toolhead part failure point

Design Analysis

By the end of our project, we were able to successfully print a line **(Figure 16)**. We achieved this by designing and manufacturing multiple parts of the system such as the end effector mount, tool head, filament routing system, and wiring **(Figure 11)**.

Overall, the tool head we designed worked effectively for our purpose of printing a line, however, we did notice a few issues arise during testing. There was a notable issue in the interfacing between the mount and toolhead where it was not as snug as it could have been, leading to some wobble. While this did not affect our line printing test, this issue could be problematic when experimenting with more complex and dynamic movements and even in cases where the tool head ends up dragging on the build plate.

Additionally, while the filament routing system worked well for the first couple of tests, running subsequent tests uncovered some issues. While the PTFE tube has little resistance when in a relatively straight path, any extreme bends make pulling the filament through the tubes extremely difficult. In the first test, the filament was able to be pulled through the tubing as it was stretched out. However, after letting the system set for an hour or two, the tube curled up and the force required to pull the filament was greater than the orbiter extruder could manage. This led to us disconnecting the tube and manually attaching the filament for subsequent tests. We aim to fix this issue in the future by looking into larger-diameter PTFE tubes allowing for more tolerance for the filament especially during bends in the tube.

One other factor that played a big part in our project was the build surface we printed on. While it was not something we were tasked to work on, it played a crucial role in our success. The print surface is called Gecko Tek **(Figure 16)** and does a good job of adhering the molten plastic to its surface. However, the build plate was mounted to a slightly uneven platform. This

means that the surface was not perfectly level in regards to the UR5e's cartesian movement plane. While the variance in height with regards to the nozzle was low enough that we were able to print a large layer height (roughly 1mm), the tool head experienced a collision with the print surface when trying smaller layer heights such as 0.2mm.

This collision occurred when we were planning out the path for the line test. While running the movement at a height of 0.2mm above the Gecko Tek platform, the nozzle began scraping and dragging onto the surface. This created torque on the printed part and snapped it along its layer line where the mount and tool head are mated together **(Figure 17)**. While this setback almost caused us to not meet our deadline, it had some unintended benefits. Since the part we printed was plastic, it had a sheer force that was much lower than the other components of the system. This means that instead of the expensive hot end breaking in half, the force was transferred to the plastic and the part failed there. Should the printed part have been made of a stronger material, there is a likely chance that the much more expensive hot end could have broken. This worked to our advantage, as the printed piece took only a few hours to rebuild, whereas a broken hot end could have delayed us for a week and caused us to miss our project deadline.

This result was also proved through the static structural deformation analysis we performed on Ansys **(Figure 18)**. This analysis was performed to see the behavior of the tool head when undergoing extreme loads to help us find potential failure points. By looking at the figure, it can be seen how the deformation begins as the part undergoes roughly 8 pounds of force, causing it to bend forward. As this continues, the part begins to break as seen in **Figure 19** where the simulation stops its deformation right before it breaks. While the Ansys analysis offers a decent view into the exact scenario, the actual tool head, being 3D printed, is weaker when

force is being applied along the layer lines. This would result in the actual part being weaker than the analysis describes even though both parts broke at similar areas both in the analysis and in real life **(Figure 17)**.

Future Work

We continue to explore additional avenues for enabling swarm applications and refining our additive manufacturing system.

One aspect being worked on is automating the slicing and path planning process for the UR5e system, where the user inputs a CAD file into a program that creates path commands for the UR5e joints and G-code for the extrusion system to print the piece successfully.

Another aspect being worked on is a robotic heating bed allowing another axis of freedom to enhance research into swarm manufacturing. This design allows for a moving bed to allow optimization for swarm manufacturing prints. By working on this aspect of the project, we can ensure consistent bed adhesion and enhance the efficiency and synchronization between multiple robotic arms while printing.

Another avenue being explored is using the Jackal UGV with the UR3 mounted on it to create an automated printed piece retrieval system. Once the UR5e system completes the printing process, the Jackal uses a multi-sensor stack to visualize, plan, and execute a movement path to the print bed, then the UR3 uses kinematic techniques to pick up the printed piece, and finally, the Jackal returns to its original position, autonomously retrieving the product.

Additionally, we are continuing to iterate and improve upon the hardware of the system. The toolhead is being redesigned with a higher flow hot end and designed with non-planar printing in mind. The extrusion system is also being improved by adding rotating rings to hold

the tubing to better manage the filament path. We are also collaborating with another team to simplify the wiring and integrate the current electronics with the UR5e's control box.

Conclusion

In conclusion, the project to develop a working model of the extrusion system for the UR5e was a success. There was an equal contribution from all team members: 33.33% from Ian, Parth, and Retvin, with mentoring and support from faculty members Dr. Sha and Ronnie Stone. We created and assembled multiple components of the system, including a tool mount, tool head attachment, hub cap for filament routing, mount for the arm, and extruder. Through the project, we found that swarm manufacturing has major implications for manufacturing using 3D printing since it enables conformal printing, which is 3D printing in non-planar geometries allowing the printing of complex shapes. It will also make manufacturing more efficient and scalable. The development of such a system is only one step in the overall progress of the manufacturing industry and it must continue to grow and innovate.

Appendix

A. References

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